

Topics in Software Engineering – Software Reengineering

By

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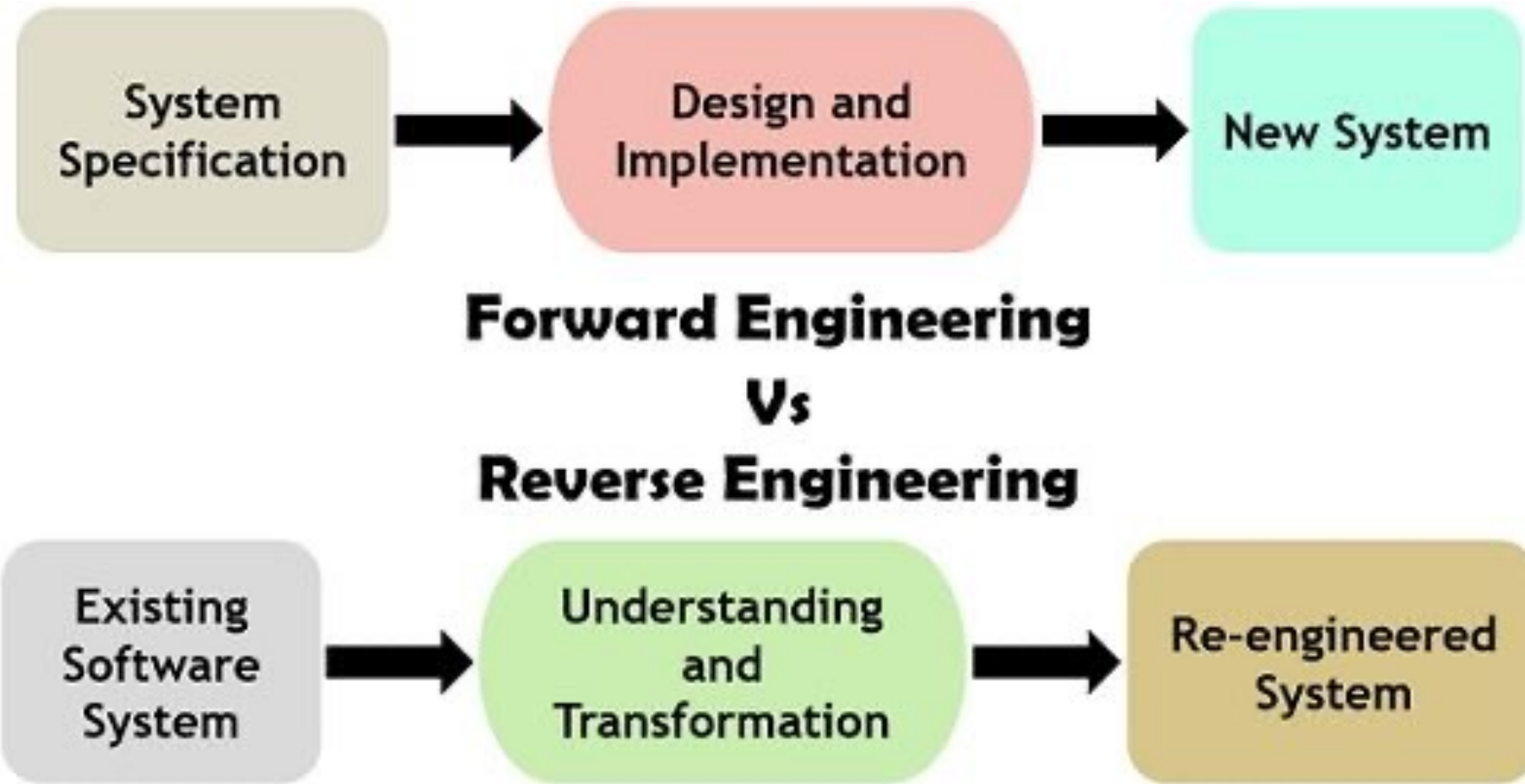
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Reverse Engineering 2

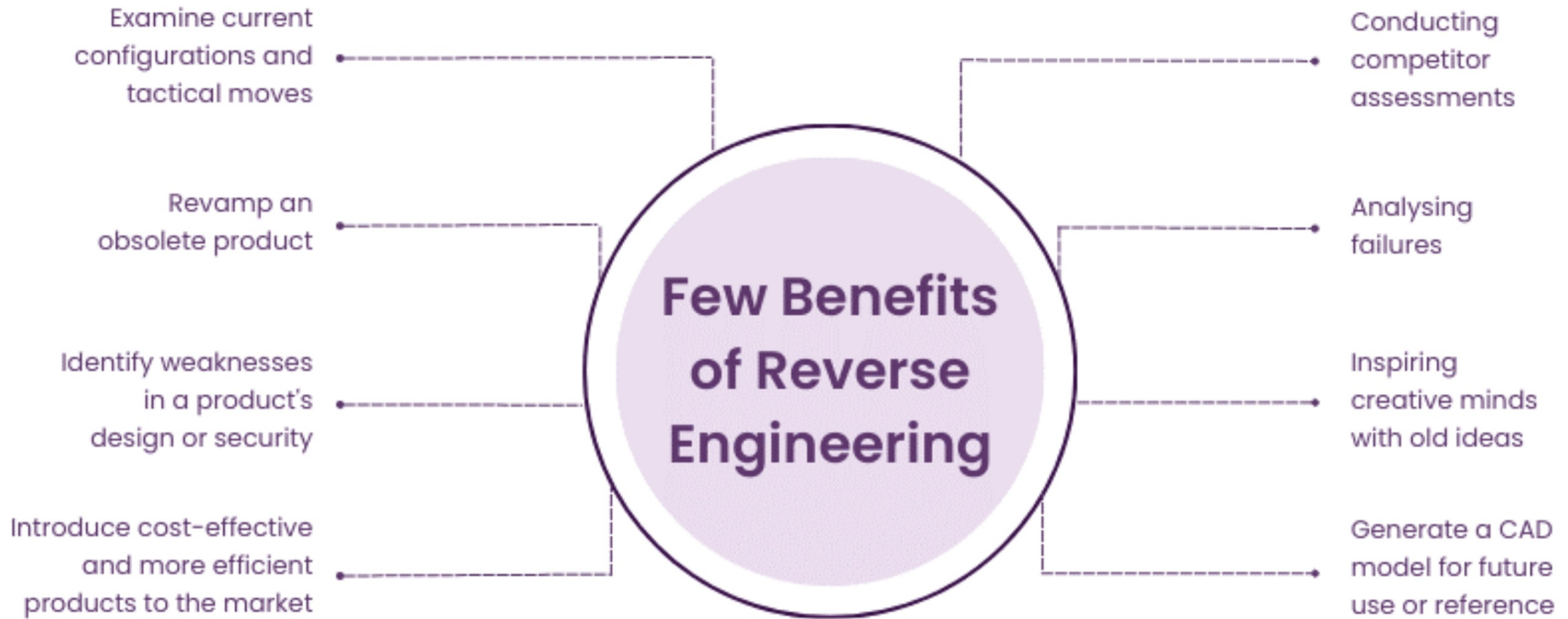


Concepts and Methods

Difference Between Forward Engineering and Reverse Engineering



Advantages of Reverse Engineering



Reverse Engineering

- “Reverse engineering is the process of analyzing a subject system to identify the system’s components and their interrelationships and create representations of the system in another form or at a higher level of abstraction”

By – Chikofsky and Cross (1990).

- In reverse engineering, **the requirements** and **the essential design, structure** and **content of the legacy system** must be recaptured.
- The effectiveness of this process will affect the success of the reengineering project.
- **Reverse engineering does not involve changes to the system or the creation of a new system.**
- It is the process of **examination without changing its overall functionality.**

Reverse Engineering Objectives

- **Cope with complexity;**

- Have a better understanding of big and complex systems,
- Automatic techniques provide a way to extract relevant information and leave out low-level details.

- **Generate alternative views;**

- Reverse engineering tools provide different views of the systems,
- It enables the designers to analyze the system from different angles.

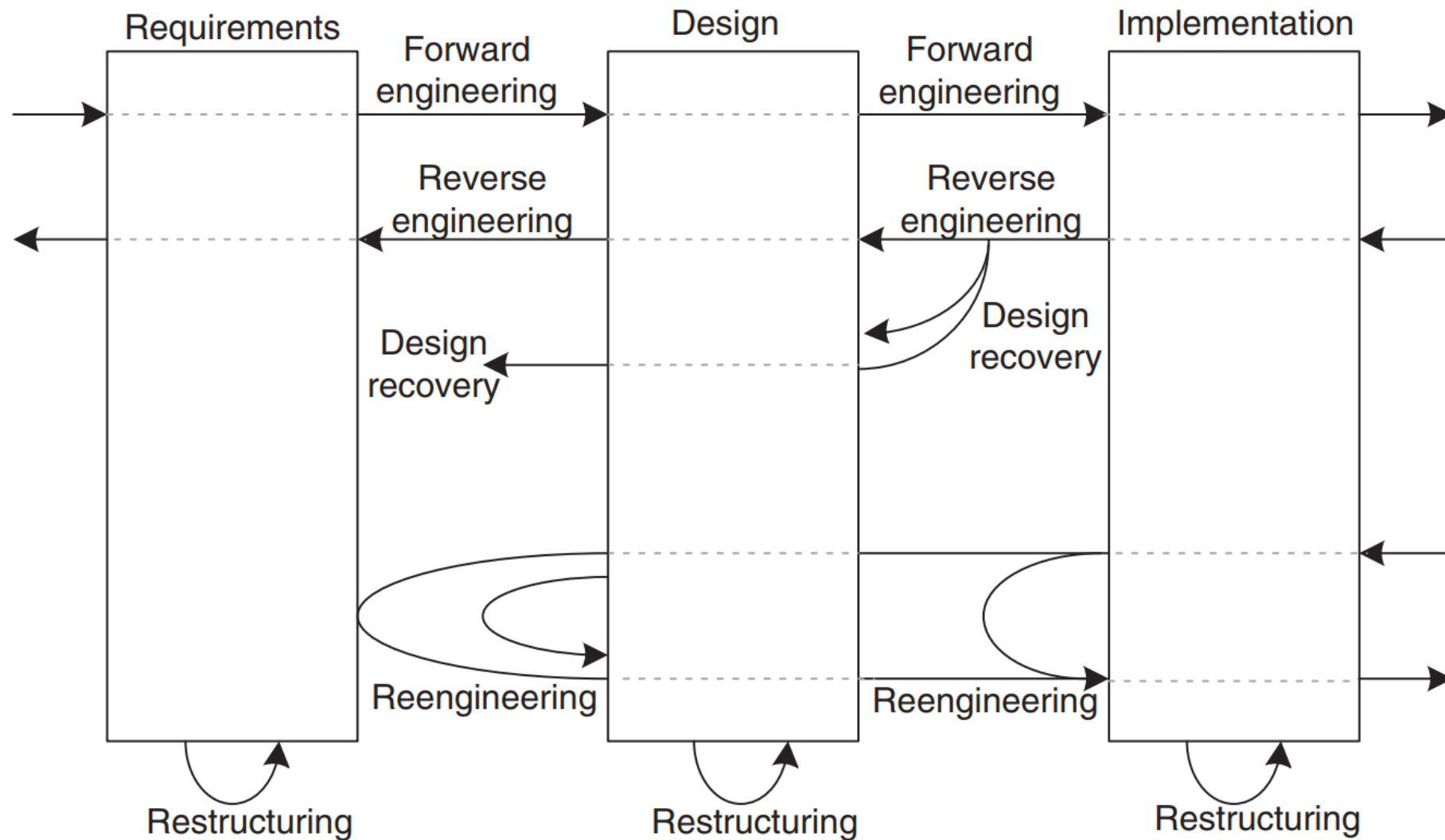
- **Recover lost information;**

- Changes made to the system are often undocumented – this enlarges the gap between the design and the implementation,
- Reverse engineering techniques retrieve the lost information.

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- **Detect side effects;**
 - Reverse engineering can help detect problems due to the effect some changes may have on the system before they result in the failure of the system.
- **Synthesize higher abstractions;**
 - Extract high-level abstract views of the system.
- **Facilitate reuse;**
 - Reverse engineering can help detect candidate system components that can be reused.

Relationship Between Reengineering And Reverse Engineering. From Reference 6. © 1990 IEEE



Factors Necessitating the Need for Reverse Engineering

- The original **programmers** have left the organization.
- The **language** of implementation has become obsolete, and the system needs to be migrated to a newer one.
- There is **insufficient documentation** of the system.
- The company acquired the system as part of a larger acquisition and lacks **access to the source code**.
- The system requires adaptations and/or **enhancements**.
- The software does not operate as expected

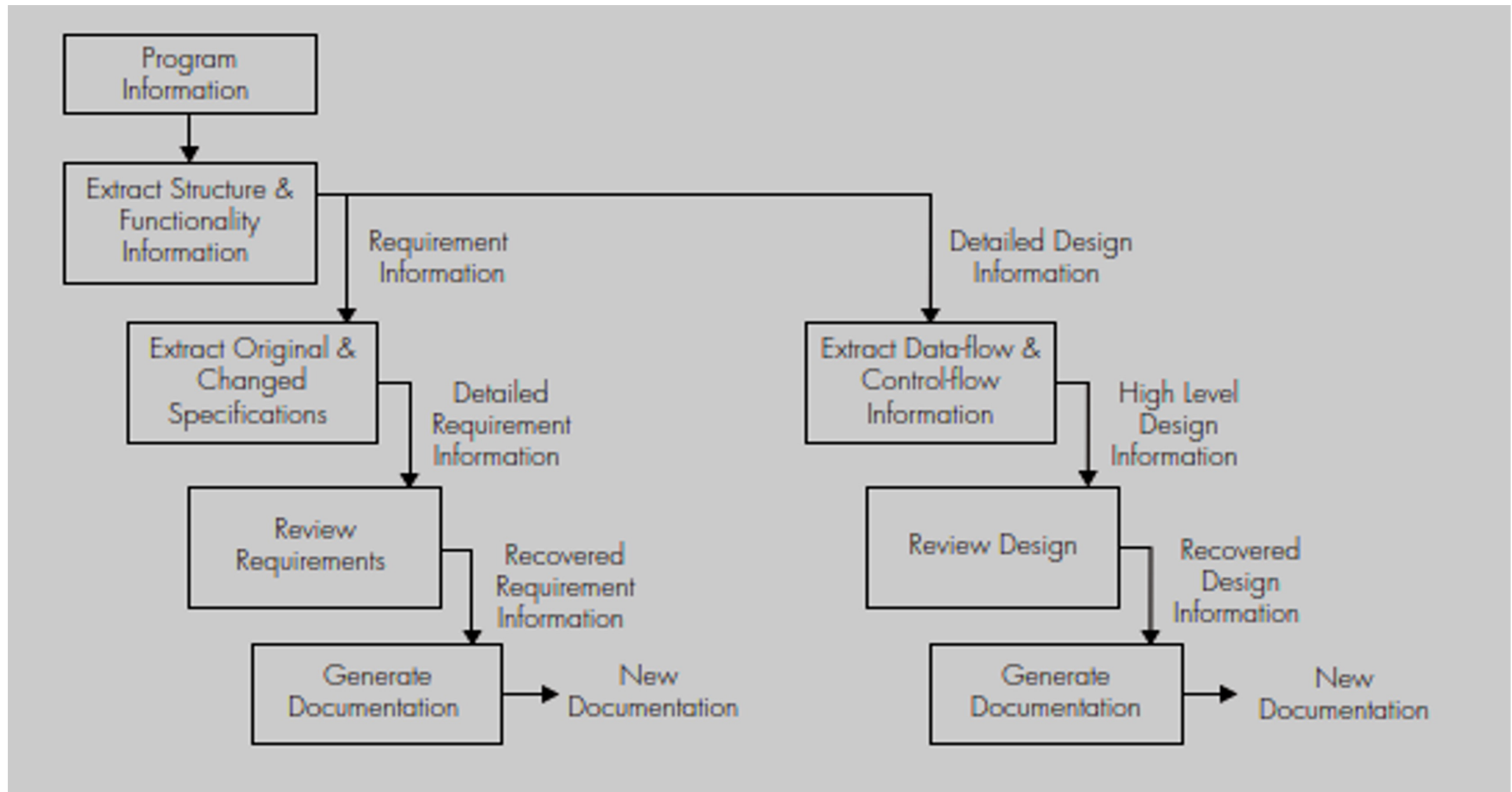
Reverse engineering applied problem areas:

- redocumenting programs [21]
- identifying reusable assets [22–25];
- discovering design architectures [7, 26–30];
- recovering design **patterns** [31, 32];
- building traceability between code and documentation [33–35];
- finding objects in procedural programs [36];
- deriving conceptual data models [37–40];
- detecting duplications and **clones** [41–44];
- cleaning up **code smells** [45];
- aspect-oriented software development [46];
- computing change impact [47];
- transforming binary code into source code [48];
- redesigning user interfaces [49–51];
- parallelizing largely sequential programs [52];
- translating a program to another language [53, 54];
- migrating data [55];
- extracting business rules [9, 56, 57];
- auditing security and **vulnerability** [59, 60]; and
- extracting protocols of network applications [61].

Reverse Engineering Process

- The reverse engineering process, begins by **extracting the requirements and detailed design information from the source code**
- A requirements document is created and a **high level design abstraction is extracted** and expressed using data-flow and control-flow diagrams.
- The recovered design is reviewed for consistency and correctness.

Reverse Engineering Procedure



Reverse Engineering Procedure

- **Collect information:**

- Collect all the possibly available information about the program.
- The sources of information include design documents, documentation for system calls and external routines and source code.
- Personnel experienced with the software should also be identified.

- **Examine information:**

- Thoroughly review the collected information.
- This step allows the people working on the recovery to become familiar with the system and its existing components.
- A plan for dissecting the program and recording the recovered information can be formulated during this stage.

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- **Extract the structure:**

- Identify the structure of the program and use this to create a set of structure charts.
- Every node in the structure chart corresponds to a particular routine called in the program.
- Hence, the chart makes a record of the calling hierarchy of the program.
- For each edge in the chart, the data passed and returned by a node must be recorded.

- **Record functionality:**

- For every node in the structure chart, record the processing done in the program routine corresponding to that node.
- A Program Description Language (PDL) can be used to show the functionality of program routines.
- For system and library routines the functionality can be described in English or in a more formal notation.

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- **Record data-flow:**

- The recovered program structure and PDL can be analyzed to identify data transformations in the software.
- These transformation steps show the data processing done in the program.
- This information is used to develop a set of hierarchical data flow diagrams that model the software.

- **Record control-flow:**

- Identify the high-level control structure of the program and use control-flow diagrams to record it.
- This refers to high-level control that affects the overall operation of the software, not to low-level processing control.

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- **Review recovered design:**

- Review the recovered design for consistency with available information and correctness. Identify any missing item of information and attempt to locate it.
- Review the design to verify that it correctly represents the program.

- **Generate documentation:**

- The final step is to generate complete design documentation.
- Information explaining the purpose of the program, program-overview, history and so on will need to be recorded.
- This information will most probably not be contained in the source code and must be recovered from other sources.

Reverse Engineering Tools

- System Monitoring Tools
 - Network activity, file access and register access
- Disassemblers
 - Translate binary code to assembly code
- Debuggers
 - It used in disassembling mode to set break points and step through program execution
- Decompilers
 - Attempt to produce high level code (e.g, c) from an executable binary code.
- Hex Editors
 - Read executing program from RAM
 - Enable the editing of running hexadecimal code

Thanks for your attention!

Any Question?



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